

From Local Roots to a More Secure Tomorrow

A staged programme using mature technology, measured evidence and modular design to prove what works in Trinidad first — then replicate only the parts that earn their right to scale.

CONCEPT DESIGN — NOT FOR CONSTRUCTION · ALL DRAWINGS NOT TO SCALE



PRIVATE SITE = PROVE

Pilot / First Commercial Campus

Edinburgh Road and Thompson Road, Carlsen Field. Approximately 3.56 acres of private family land. Build Wing 0 (existing structure, approximately 279 m² subject to survey verification), compact Spine, Wing 1, packhouse, utility yard. Validate climate, yield, uptime, water, energy and maintenance costs under local conditions. Produce first commercial sales. Train people and refine operating procedures.

- WING 0
- SPINE
- WING 1
- PACK + COLD
- UTILITY YARD

STATE SITE = SCALE (ONLY IF GATES PASS)

Conditional Full-Scale Campus

Opposite state land (approximately 16 acres) south of Thompson Road. Tenure unverified — state land allocation is conditional, subject to Commissioner of State Lands approval, EMA Certificate of Environmental Clearance, DevelopTT planning, and other statutory requirements. Replicate proven modules — not untested ambition. Preserve failure isolation and maintainability. Expand only against demonstrated demand, confirmed finance, secured tenure and obtained approvals.

- WINGS A–D
- CENTRAL SPINE
- ADMIN + TRAINING
- STORMWATER
- FUTURE RESERVE

~3.56

acres (private)

Pilot site, family land

~16

acres (conditional)

State land, unverified tenure

2+R

levels

Spine: ground, upper, roof plant

5

gates

Evidence-based decision points

Carlsen Field is not a bet on an expensive futuristic farm. It is a staged programme that uses mature technology, measured evidence and modular design to prove what works in Trinidad first — then replicate only the parts that earn their right to scale.

Every drawing in this concept set tells that same story.

One Design Basis for Both Documents

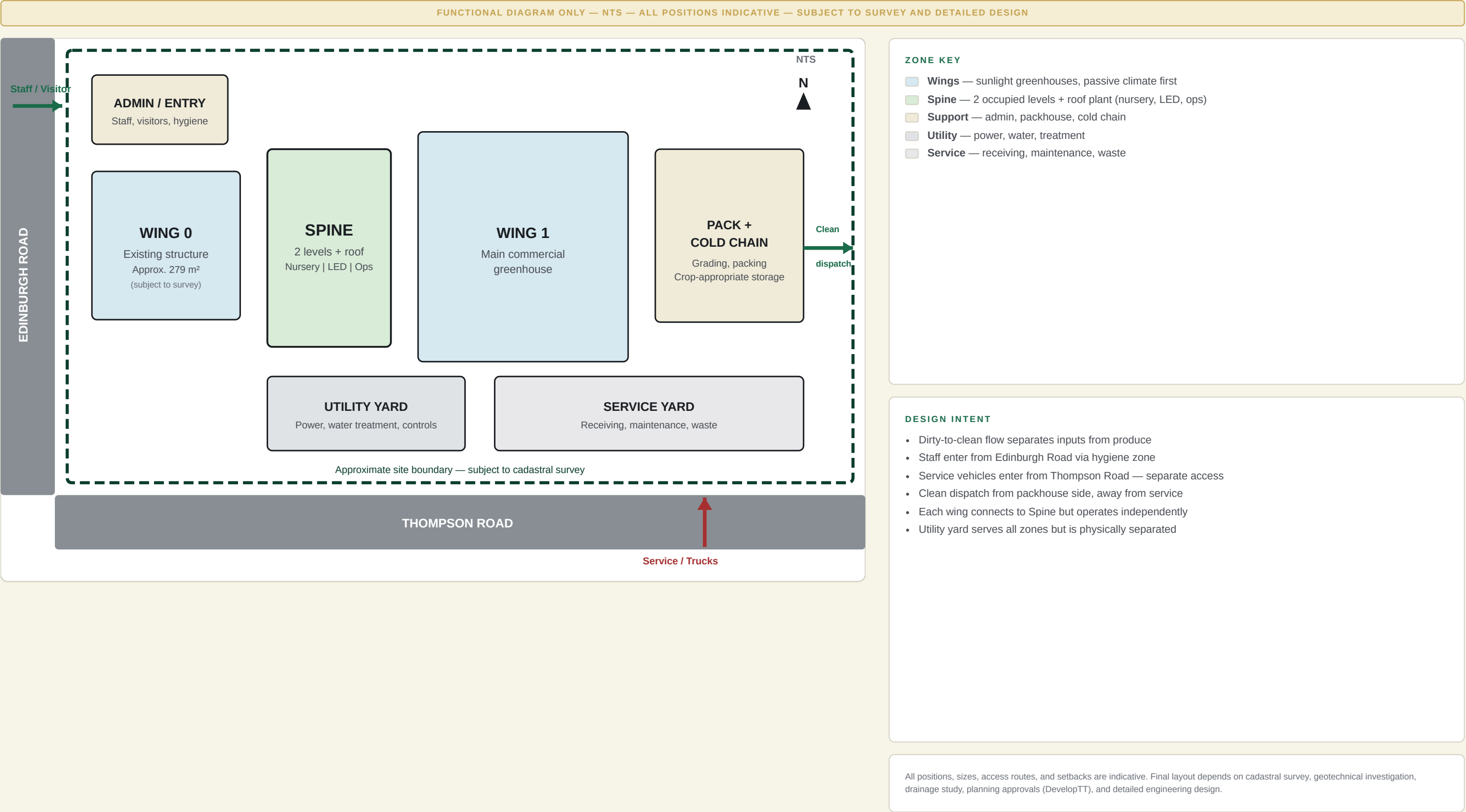
Every concept in this programme rests on the same assumptions. Where an assumption has not yet been verified, it is flagged as such. No claimed compliance, no committed values — concept status throughout.

Category	Design Basis Statement	Status / Qualification	Verification Required
Site — Pilot	Private land, approximately 3.56 acres, Edinburgh Rd / Thompson Rd, Carlsen Field	Deed held; exact boundaries and levels require survey	Cadastral survey, boundary confirmation, topographic levels
Site — Scale	State land approximately 16 acres opposite Thompson Road	Conditional — tenure not secured, area approximate	Commissioner of State Lands application; boundary survey; EMA CEC
Climate	Tropical maritime; ambient typically 24–34 °C; high humidity year-round	General knowledge; site-specific psychrometric data not yet collected	On-site weather station; minimum 12 months logged data before HVAC sizing
Power Supply	T&TEC grid as primary; backup generator with ATS; UPS for controls	T&TEC supply availability to be confirmed; generator and UPS sizes TBD	T&TEC load enquiry; Govt Electrical Inspectorate for installation certification
Water Supply	WASA mains, rainwater harvesting, potential well — sources to be tested	WASA availability not confirmed; well yield and quality unknown	WASA connection enquiry; water quality testing; well yield test if applicable
Structure	Wing 0: existing structure to be assessed and retrofitted. New structures concept only.	No structural design completed; no assumed dimensions or loadings	Structural assessment of Wing 0; geotechnical investigation; structural engineer
Drainage	Site drainage and stormwater management required for Trinidad rainfall	No drainage design; catchment and discharge routing unknown	Hydrological study; EMA requirements; outfall identification
Growing System	Recirculating hydroponics; nutrient mixing per module; no shared untreated return	System type to be confirmed through trials; specific equipment TBD	Trial crops in Wing 0; nutrient management validation; local water analysis
Post-Harvest	Packhouse with grading, packing, cold storage — crop-appropriate temperatures	Cold chain temperatures vary by crop (basil is chilling-sensitive at below approximately 12 °C; tomato storage varies by maturity stage)	UC Davis produce fact sheets; crop-specific trials; cold room sizing after demand
Regulatory	Planning via DevelopTT; EMA CEC; T&TEC; Govt Electrical Inspectorate; WASA; fire; food safety	No applications submitted; requirements to be confirmed with each authority	Pre-application enquiries to each authority; legal review of requirements
Seismic / Wind	Trinidad is in a seismically active region with hurricane-season wind exposure	No assumed zone values, wind speeds, or structural design parameters	Structural engineer to determine applicable codes and design parameters
Commercial	Market demand for target crops to be validated before production commitment	Buyer interest to be confirmed; pricing assumptions are preliminary	Buyer engagement; price benchmarking; demand validation at Stage 0

S03 / PILOT SITE CONCEPT ARRANGEMENT

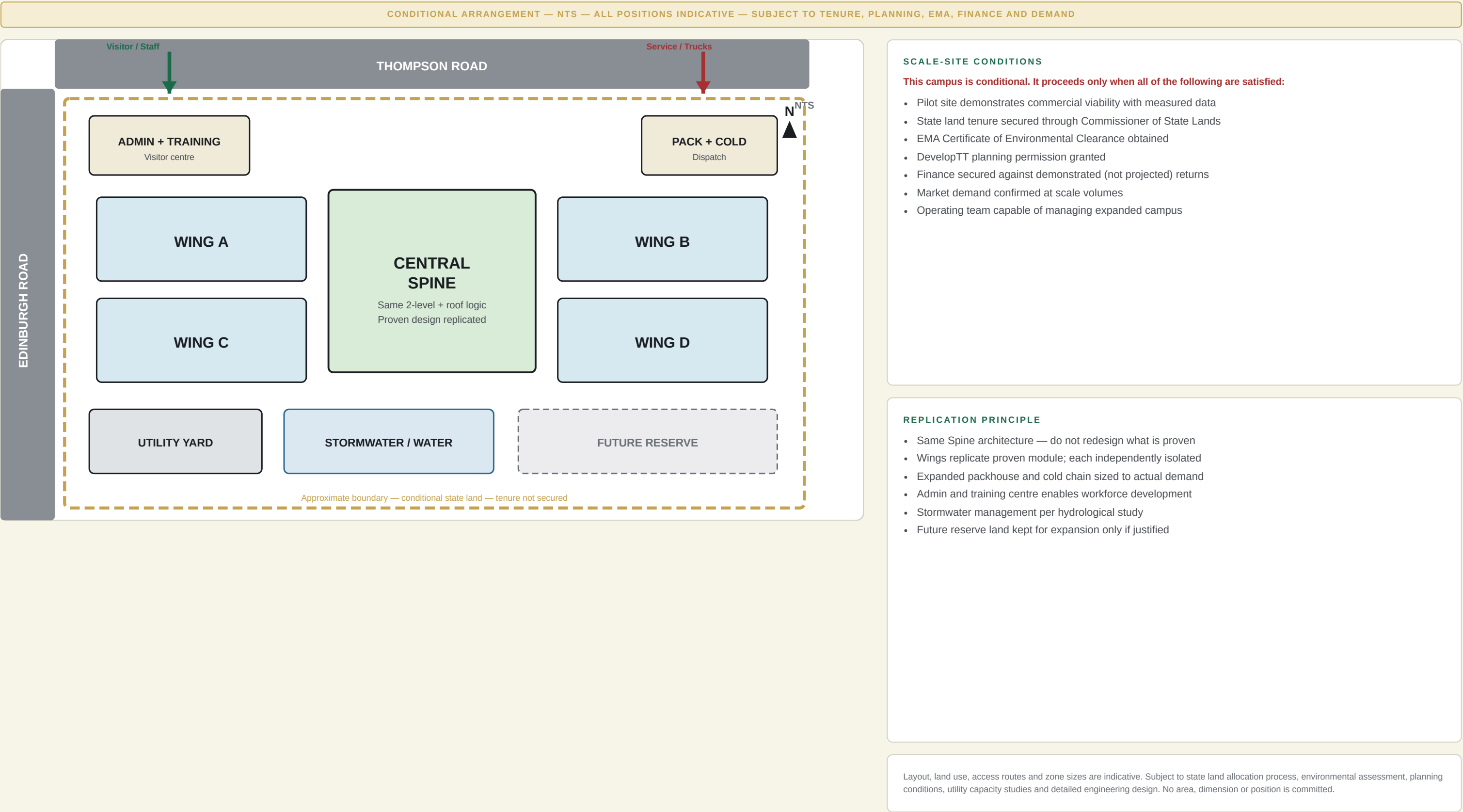
Private Pilot Site — Functional Map

Approximately 3.56 acres, Edinburgh Road / Thompson Road, Carlsen Field. This is a concept-level functional arrangement showing zone relationships — not a dimensioned site plan. All positions, sizes and access routes are indicative and subject to survey, detailed design and authority approvals.



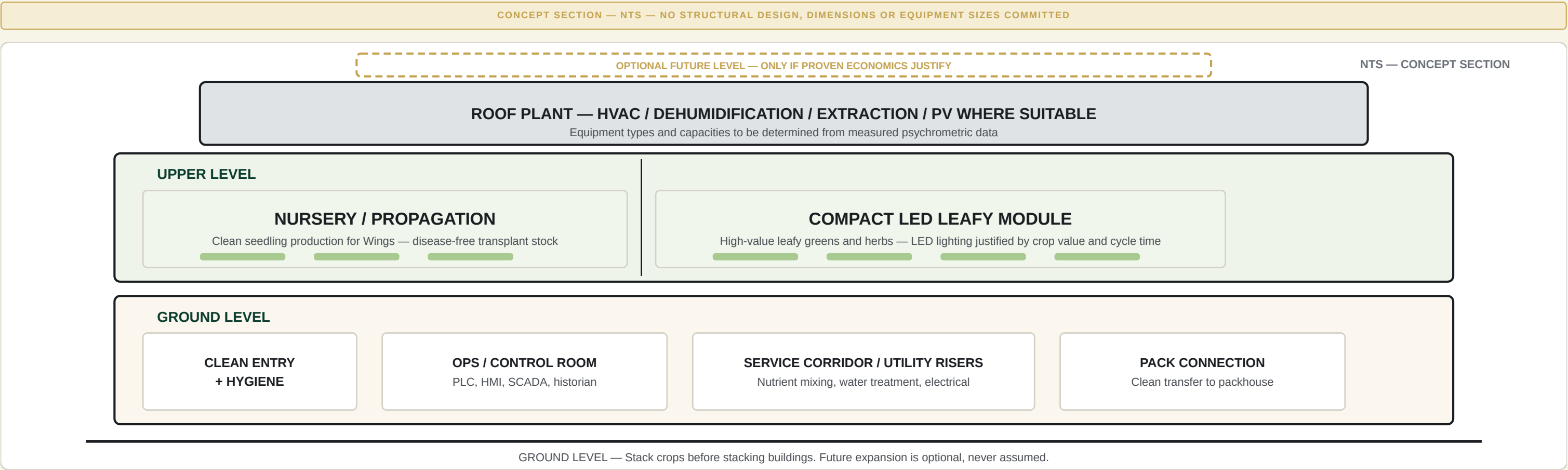
Scale Site — Only If All Gates Pass

Approximately 16 acres opposite Thompson Road. State land — tenure not secured, area approximate. This arrangement is conditional on: pilot operating data, Commissioner of State Lands approval, EMA CEC, DevelopTT planning permission, secured finance, and confirmed market demand. Replicate proven modules, not untested ambition.



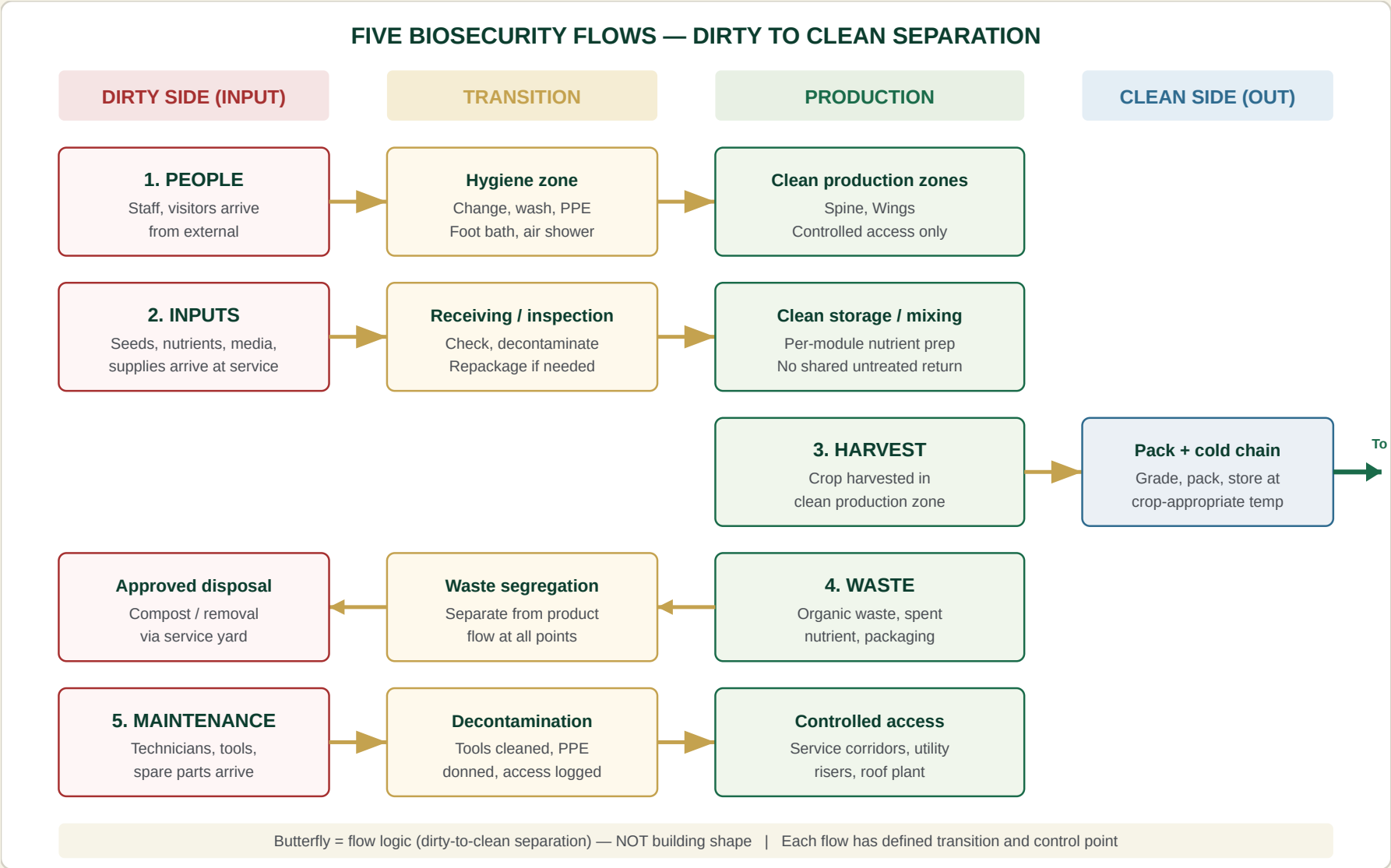
Two Occupied Levels + Roof Plant

The Spine is the technical core of the campus. It houses the functions that require the most control — nursery propagation, compact LED leafy production, operations and controls — stacked vertically to reduce footprint. Roof level carries HVAC, dehumidification and extraction plant. This is a concept section — no dimensions, structural sizes or equipment specifications are committed.



Five Flows — The Butterfly Logic

The Butterfly is a process flow concept, not a building shape. It describes how five distinct flow categories move through the campus from dirty side to clean side, with controlled transitions. Every flow is managed to prevent contamination crossing from one zone to another.



1. People Flow

All personnel transition through hygiene zone before entering production. Visitors restricted. Movement logged. Clean-side clothing only in production areas.

2. Input Flow

All materials received at service yard (dirty side). Inspected, decontaminated and repackaged before transfer to clean storage. No direct path from external to production.

3. Harvest Flow

Product moves from production to packhouse through clean transfer. Graded, packed and stored at crop-appropriate temperatures. Basil above approximately 12 °C (chilling-sensitive). Dispatch from clean side only.

4. Waste Flow

Organic waste, spent nutrient solution, and packaging routed out of production through dedicated waste path. Never crosses product flow. Exits via service yard for approved disposal or composting.

5. Maintenance Flow

Technicians and tools enter through decontamination. Access to service corridors, utility risers and roof plant via controlled routes. Maintenance activity logged. Tools cleaned on entry and exit.

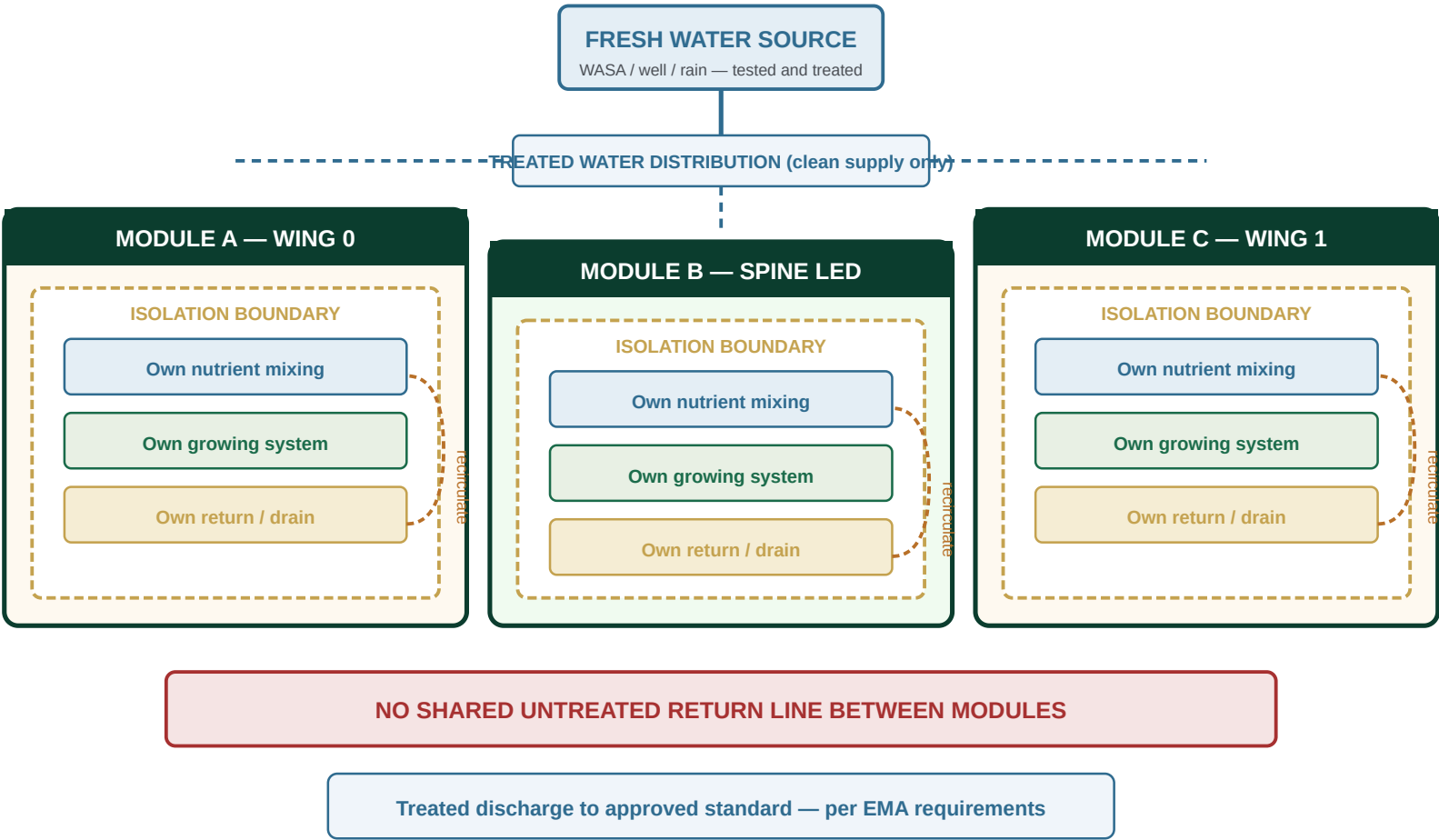
The five flows are a design principle. Detailed SOPs, access protocols and physical barriers will be developed during detailed design with reference to food safety and biosecurity best practice.

Keep Failure Inside One Module

Each production module (wing or Spine zone) has its own nutrient mixing, crop return and isolation boundary. No shared untreated return lines. If a disease, contamination or system failure occurs in one module, it does not propagate to others. This is the most important design principle in the entire programme.

NUTRIENT ISOLATION SCHEMATIC — EACH MODULE INDEPENDENT

Concept diagram — NTS — no pipe sizes, tank volumes or equipment specifications committed



WHY ISOLATION MATTERS

- **Disease containment:** Root-zone pathogens (Pythium, Fusarium) cannot spread between modules through shared nutrient lines
- **Contamination control:** A chemical error in one module's mixing does not poison other crops
- **Failure isolation:** A pump failure, leak or sensor fault affects only one module
- **Independent diagnostics:** Each module's nutrient, pH, EC and flow data is self-contained — easier to identify and fix problems
- **Staged commissioning:** Modules can be commissioned, tested and adjusted independently

WHAT EACH MODULE CONTAINS

- Own nutrient stock tanks and dosing (A/B concentrate + pH)
- Own mixing and delivery system
- Own growing system (NFT, DWC, drip — type per crop)
- Own return collection and recirculation
- Own drain-to-waste or treated discharge path
- Own sub-metering: water, nutrient, energy, climate

Fresh water is the only shared input. Everything downstream of the mixing point is module-specific. This is non-negotiable.

S08 / PHASING AND INVESTMENT GATES

Stages 0–4 — No Jump from Idea to Full Build

Gate-based programme. No calendar promises. Proceed only when gate criteria are met with measured evidence. Each stage has explicit pass/fail criteria; the outcome at every gate is Go, Redesign, Defer, or No-Go.

0

Stage 0 — Measure and Verify

PRE-BUILD

Activities:

- Cadastral survey, geotechnical investigation, drainage study
- Confirm planning path (DevelopTT), utility connections (T&TEC, WASA)
- Water quality testing, source verification
- On-site psychrometric data collection (minimum 12 months)
- Market research: buyers, pricing, demand confirmation

Gate 0 Criteria:

- Site boundaries and levels confirmed
- Planning pathway clear
- Water source viable
- Climate data sufficient for HVAC concept
- At least one buyer engaged
- Investment case supportable

1

Stage 1 — Wing 0 Proving Cell

LOW CAPITAL

Activities:

- Assess and retrofit existing structure (approximately 279 m², subject to survey)
- Install full monitoring: climate, water quality, energy metering
- Test crops, climate management, operating procedures
- Train first operators; build maintenance knowledge

Gate 1 Criteria:

- Crop yield data from local conditions
- Energy and water consumption measured
- Operating cost baseline established
- Team trained and competent
- System reliability demonstrated

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Stage 2 — Pilot / First Commercial Campus

PRIMARY INVESTMENT

Activities:

- Build compact Spine (2 levels + roof) and Wing 1
- Packhouse and cold chain facility
- Full utilities: power, water, fertigation, controls
- First commercial sales and market integration

Gate 2 Criteria:

- Stable production at target quality
- Buyer relationships established
- Unit economics measured (not projected)
- Uptime and reliability data
- Maintenance costs tracked

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Stage 3 — Controlled Pilot Expansion

IF JUSTIFIED

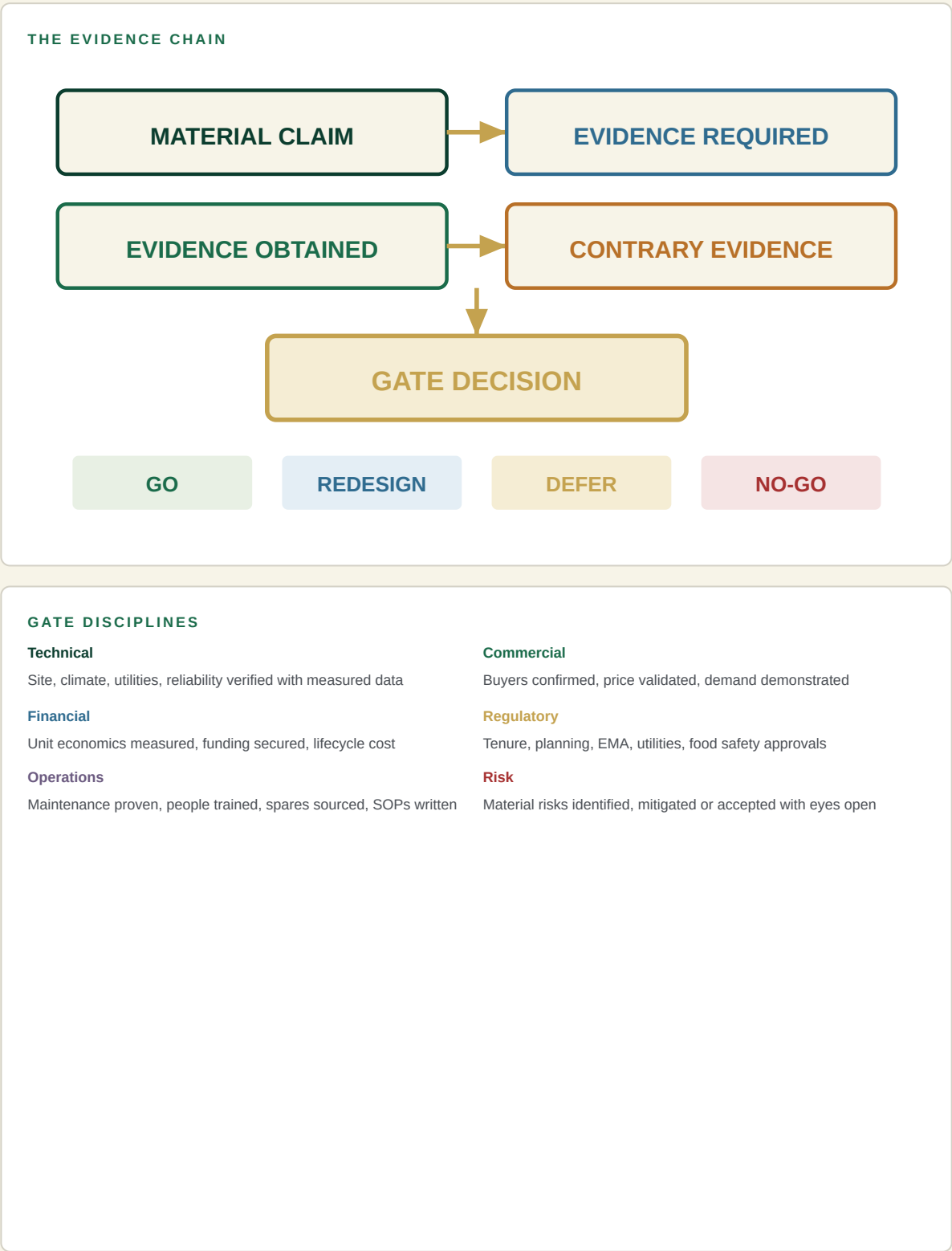
Additional modules on pilot site only if demand and economics justify. No second commercial wing assumed in base case. Expand operating team and training programme. Decision driven by measured margins, not projections.

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Stage 4 — Conditional Full-Scale Campus

SUBJECT TO ALL GATES

State land allocation pursued only with proven operating data. Requires: tenure (Commissioner of State Lands) + planning (DevelopTT) + environmental (EMA CEC) + finance + demand. Replicate proven modules; four Wings (A–D) around central Spine. Regional replication potential from demonstrated model.



Choose Crops Through Buyers and Measured Margins

Crop selection is not a design decision made in advance — it is a commercial decision driven by buyer requirements, local market pricing, measured production costs and proven yield. The architecture accommodates a range of crops; the market determines which ones earn their place.

SPINE CROPS — LED / HIGH-CONTROL ENVIRONMENT

Nursery and High-Value Leafy Production

Category	Candidate Crops	Rationale
Seedlings	Tomato, pepper, lettuce transplants	Clean propagation; disease-free stock for Wings
Leafy Greens	Lettuce varieties, pak choi, kale	Short cycle; benefits from climate control; local demand
Herbs	Basil, cilantro, mint, chives	Year-round supply; hotel/restaurant demand. Note: basil is chilling-sensitive — storage above approximately 12 °C
Microgreens	Sunflower, pea shoots, radish	High value per area; fast turnover; niche market potential

WING CROPS — SUNLIGHT / PROTECTED GREENHOUSE

Commercial Greenhouse Production

Category	Candidate Crops	Rationale
Tomatoes	Indeterminate varieties	High local demand; import substitution; year-round premium. Post-harvest temperature varies by maturity.
Sweet Peppers	Bell, seasoning peppers	Import substitution; consistent quality; existing market
Cucumbers	English cucumber	Fast cycle; high yield per area; market familiar
Leafy Greens	Lettuce, bok choy, callaloo	Low energy; uses natural light; complements Spine output

CROP SELECTION PRINCIPLE

The buyer decides what we grow. Measured margins decide if it is viable.

- No crop is committed until buyer demand and pricing are confirmed
- Production cost must be measured in Wing 0 / Stage 1 before commercial commitment
- Yield data from local conditions, not imported assumptions
- Post-harvest handling is crop-specific — no universal cold chain temperature
- Crop mix may change over time as market and margins evolve

MARKET FLOW



Crop-appropriate temperatures at each stage. No universal 4 °C cold chain.

TARGET MARKET CHANNELS

Supermarkets

PriceSmart, Massy, local chains. Consistent volume and quality required.

Institutional

Schools, hospitals, catering. Volume at competitive pricing.

Hotels and Restaurants

Premium herbs and greens. Year-round availability is the value.

Direct / Market

Community access. Brand building. Feedback loop on quality.

All crop candidates are subject to trial in Wing 0 / Stage 1. No production commitment is made until measured yield, quality and cost data confirm commercial viability under Carlsen Field conditions.

What the Investment Must Survive

Every risk is addressed by the gate-based approach. No commitment without evidence. The design philosophy: reduce exposure to unresolved material assumptions before capital is committed. These risks are acknowledged, not dismissed.

Energy Cost and Grid Reliability

HIGH IMPACT

Risk: T&TEC tariff changes or extended outages could undermine operating economics. Generator fuel costs are volatile.

Mitigation: Passive-first Wing design minimises energy demand. Generator + UPS provides backup. Sub-metering tracks every kWh. Solar PV reduces grid dependency. Survival Mode protects critical loads. Actual energy costs measured in Stage 1 before scaling.

State Land Tenure

HIGH IMPACT

Risk: Full-scale campus depends on state land allocation through Commissioner of State Lands — process may be slow, conditional or unsuccessful.

Mitigation: Pilot is entirely on private land. State land is Stage 4 only. The pilot must prove viability independently. No capital is at risk from tenure uncertainty.

Market Demand and Pricing

MEDIUM IMPACT

Risk: Local market may not absorb production at economically viable prices. Competition from imports or other producers.

Mitigation: Market validation is a Stage 0 gate requirement. First commercial sales at Stage 2 before scaling. Multiple channels. Scale only when demand is measured, not assumed.

Climate Performance Gap

MEDIUM IMPACT

Risk: Imported CEA assumptions may not hold under Trinidad's heat, humidity and rainfall. Equipment sized from temperate-climate data may underperform.

Mitigation: Wing 0 proving cell tests everything locally before building Wing 1. On-site psychrometric data drives HVAC sizing — no imported specifications used unchecked. UWI research supports local evidence base.

Supply Chain for Spares and Consumables

MEDIUM IMPACT

Risk: Specialised components may be hard to source in Trinidad. Lead times and FX costs for imported parts.

Mitigation: Design uses mature, widely available technology. Manual fallback for critical functions. No proprietary lock-in. Local technician maintainability is a design requirement.

Regulatory and Approval Risk

MEDIUM IMPACT

Risk: Planning (DevelopTT), environmental (EMA CEC), utility approvals, and food safety certification timelines are uncertain.

Mitigation: Pre-application enquiries at Stage 0 before capital commitment. No construction starts without required approvals. Regulatory pathway is a gate criterion.

Water Source Reliability

LOW-MEDIUM IMPACT

Risk: WASA supply interruptions; well yield may be insufficient or water quality unsuitable.

Mitigation: Water testing and source verification is Stage 0 gate. On-site storage sized for supply interruptions. Rainwater harvesting supplements supply. Recirculating hydroponics minimises total consumption.

Workforce and Skills

LOW-MEDIUM IMPACT

Risk: Skilled greenhouse operators, maintenance technicians and data-literate staff may be scarce locally.

Mitigation: Training built into the programme from Stage 1. Wing 0 is the training ground. Standard operating procedures developed and refined before scaling. Technology chosen for local maintainability.

The gate-based approach does not eliminate risk. It ensures that each risk is identified, measured where possible, and mitigated before the capital associated with it is committed.

Who to Talk to and What to Seek

Different stakeholders need different emphasis. Same programme, same evidence standard — tailored message. This page maps who is involved and what each conversation should address.

Government / Ministry of Agriculture What They Care About - Reducing food import dependency and FX drain - National food security and resilience - Job creation and skills development - Visible, replicable success story What to Seek Policy alignment confirmation. Access to agricultural incentives. Introduction to state land allocation process. Support for SIDS food security agenda.	Commissioner of State Lands / DevelopTT What They Care About - Productive use of state land - Proven operator, not speculative proposal - Environmental compliance and community benefit What to Seek State land allocation pathway. Planning permission requirements and timeline. Pre-application guidance. Understanding of conditions and obligations.	Financial Institutions / Investors What They Care About - Clear business model and unit economics - Risk mitigation through staged investment - Proven technology, not experimental - Collateral and exit strategy What to Seek Staged financing aligned to gate outcomes. Agricultural development loans or incentives. Investment structured around measured (not projected) returns.
EMA / Environmental What They Care About - Environmental impact assessment - Water discharge and drainage compliance - Waste management plan - Noise, dust, traffic during construction What to Seek Certificate of Environmental Clearance (CEC) requirements. Pre-application guidance. Understanding of conditions for agricultural facilities in this area.	T&TEC / Govt Electrical Inspectorate T&TEC (Supply) - Load requirements and connection capacity - Tariff structure and phased demand growth Govt Electrical Inspectorate (Certification) - Installation certification and inspection - Generator and PV interconnection requirements What to Seek Load enquiry response. Tariff confirmation. Installation certification pathway.	WASA / Water Authority What They Care About - Water demand and connection capacity - Discharge compliance - Responsible use and metering What to Seek Connection availability and capacity. Water quality data for source water. Discharge requirements. Metering arrangements.
Technical Partners / Engineers What They Care About - Design that works in tropical conditions - Maintainability with local technicians - Clear specifications from measured data What to Seek Structural, mechanical, electrical engineering services. Local experience with agricultural and industrial buildings. Willingness to design from measured data, not imported assumptions.	Local Community / Carlsen Field What They Care About - Employment and training opportunities - Environmental impact, traffic, noise - Access to fresh produce What to Seek Community support and understanding. Local recruitment pipeline. Feedback on concerns. This is an agricultural campus built responsibly in the community, not a factory.	Research / UWI / Knowledge Partners What They Care About - Applied research opportunities in tropical CEA - Data sharing and publication - Student projects and training What to Seek Research collaboration on climate performance, crop trials, energy optimisation. Access to existing tropical greenhouse research (Suraj, Ekwue and Birch; Graamans et al.). Student placement programme.

What This Document Is — and Is Not

This document is a concept-level stakeholder visual set. It describes design intent, programme structure and the approach to risk. It is not a detailed design, a construction document, an engineering specification or a compliance submission. Every statement is at concept status. Nothing in this document should be read as a commitment, a guarantee, or a claim of compliance.

EVIDENCE BOUNDARY

This Document Is	This Document Is Not
A concept-level description of programme intent	A detailed design or engineering specification
A framework for staged, evidence-based investment	A financial projection or investment prospectus
Functional arrangements showing zone relationships	Dimensioned site plans or construction drawings
A map of risks acknowledged and mitigation approaches	A claim that risks are eliminated or managed
A basis for stakeholder conversations	A regulatory submission or compliance document
Concept status throughout — no item is "designed"	Ready for procurement, tendering or construction

KEY REFERENCES

- REGULATORY — ENVIRONMENT: Environmental Management Authority (EMA) — Certificate of Environmental Clearance (CEC) requirements for agricultural development projects in Trinidad and Tobago
- REGULATORY — PLANNING: DevelopTT (formerly Town and Country Planning Division) — planning permission requirements and application process
- REGULATORY — POWER SUPPLY: Trinidad and Tobago Electricity Commission (T&TEC) — tariff schedules, connection requirements, load enquiry process
- REGULATORY — ELECTRICAL CERTIFICATION: Ministry of Public Utilities, Government Electrical Inspectorate — installation inspection and certification requirements
- REGULATORY — WATER: Water and Sewerage Authority (WASA) — connection, supply, discharge and metering requirements
- REGULATORY — LAND: Commissioner of State Lands — state land allocation process (for conditional scale site)
- POST-HARVEST: UC Davis Postharvest Technology Center — Produce Fact Sheets for crop-specific temperature, humidity and storage guidance
- LOCAL RESEARCH: Suraj, Ekwue and Birch (2018) — active climate control in Trinidad and Tobago greenhouse conditions
- CEA ENERGY: Graamans et al. (2017, 2018) — energy balance comparison of plant factories and greenhouses; implications for tropical operation
- REGIONAL RESEARCH: UWI (University of the West Indies) — CEA and greenhouse research programmes relevant to Caribbean conditions

WHAT MUST HAPPEN BEFORE CONSTRUCTION

- Cadastral survey and boundary confirmation
- Geotechnical investigation
- Hydrological study and drainage design
- Structural assessment of Wing 0 (existing structure)
- Minimum 12 months on-site climate data collection
- Water source testing and quality analysis
- T&TEC load enquiry and connection confirmation
- Govt Electrical Inspectorate requirements confirmed
- WASA connection and discharge enquiry
- DevelopTT planning pre-application and submission
- EMA CEC application (for scale site)
- Structural engineering design (new structures)
- M&E engineering design (HVAC, electrical, plumbing)
- Market validation and buyer engagement
- Financial model based on measured data
- Legal review of all agreements and obligations

SIX THINGS EVERY STAKEHOLDER SHOULD KNOW

- Food security starts with evidence, not ambition.** Every material assumption is measured and verified before capital is committed.
- Private site proves, state site scales.** No taxpayer money at risk during the proving phase.
- Mature technology, smarter architecture.** The innovation is in the system design, not in unproven tech.
- Designed for Trinidad.** Heat, humidity, grid reality, local maintenance capacity — all considered.
- Jobs and capability.** Operator, maintenance, data and management skills built into the programme.
- The value is replicability.** A proven module that can be repeated responsibly.

Use mature, serviceable technology in a smarter architecture.
Complexity must earn its place.

Carlsen Field / CEA Programme
Christopher Joseph · September 2026

MEASURE › VERIFY › PROVE › SCALE

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